Requirements

Each category of requirements is defined by the IEEE software requirements specification guide: https://ieeexplore.ieee.org/document/278253 (and additional guidance from https://www.perforce.com/blog/alm/how-write-software-requirements-specification-srs-document).
From this, we created a table for each category type, and all of their properties and relations, which we then elicited requirements into, as we felt this was the clearest way to display all this information and allow iterative updates.

Requirements were elicited in several stages:
1. Initial user requirements were derived directly from the Product Brief.
This gave us a general gist of what the game is that we are developing, allowing us to start thinking about design and implementation.
2. Full user requirements were determined by meeting with the customer to fully clarify what the idea of the project is and their thoughts on specifics.
3.Functional & Quality requirements were then determined for each user requirement, ensuring each relates to customer intention, so we don't waste time developing unneeded features.
Any functional requirements we thought we needed but didn't have a user requirement were brought up for discussion, and either taken out or added a user requirement that we had missed out (such as making a user requirement for the player character, which isn't gone into detail in the product brief).
4.The requirements were used and updated accordingly over the course of the project.

The user requirements relate to the customers intention and by doing it this way, we ensure that the functional and quality requirements all relate to user requirements.
Since the process was iterative, the requirements can be gone over and refined as the project develops.
We hence presented them in a table like this to show correspondence between requirements and allow easy updating, as well as having each property of requirements clear as given in the IEEE guide, such as the priority of user requirements.

User Requirements

ID	Description	Priority
UR_PLAYERCHARACTER	The game will feature a player character that the user will be able to control.	Must
UR_GAMETIME	The game will last 5 minutes	Must
UR_PAUSING	The game should be able to be paused at any time & will start paused.	Must
UR_ITEMS	There will be at least 5 initial inventory items &There will be at least 5 map pickup items	Must
UR_GEESE	There must be a flock of geese which persistently block/attack the player.	Must
UR_WINCONDITION	There must be a way for the player to win before the 5 minute timer ends by dispersing enough geese and reaching the end point.	Must
UR_EVENTS	There will be at least 5 hidden events which the player can trigger. They must affect score in some way.	Must
UR_SCORE	There must be a score given upon ending a game. It increases if: -You diffuse the situation as fast as possible -You use items smartly -You manage hidden events correctly	Must
UR_DIFFICULTIES	There should be multiple selectable difficulties for each game. The default should be easy. Increasing difficulties should make the game harder and can restrict ability to pause.	Could
UR_ACCESSIBILITY	For any useful information the player receives, it should be shown in various forms, I.e audio, visually, typed text, etc	Must
UR_NOVIOLENCE	Any depictions of violence either visual or otherwise	Must

	must be replaced with whimsical / nonviolent alternatives.	
UR_NOCOPYRIGHTINFRINGEMENT	All assets or resources used in the production or in the final game must be licensed appropriately to prevent a lawsuit.	Must
UR_LOWDIFF	The final game must be very easy to understand and play, and extremely forgiving, i.e the win condition must be achievable by practically all players.	Must
UR_TUTORIAL	Through either implicit or explicit means, the game mechanics should be taught to the player, the specific implementation is discretionary, but a tutorial, images, is advised	MUST
UR_LIFEBAR	On the screen there should be some indicator of how close the user is to winning/losing.	Could
UR_GEESEDISPERSAL	There should be a way for the player to hinder the geese and stop their pursuit.	Must
UR_MAP	The game will feature a map resembling a university campus where the player character is present.	Must

Functional Requirements

ID	Description	Relevant User requirements
FR_TIMER	The 5 minute timer for game end as a digital clock should be on clear display to the player at all times and provide audio cues for when time is nearly up.	UR_GAMETIME, UR_ACCESSIBILITY
FR_PAUSEMENU	Upon pressing a button, the game should enter a pause menu. The game state should be completely visible while paused. It should have a resume, restart, tutorial and quit button.	UR_PAUSING, UR_ACCESSIBILITY
FR_PAUSING	the game should temporarily stop when triggered by specified events, allowing time for information to be given to the user.	UR_PAUSING UR_ACCESSIBILITY
FR_ITEMS	There should be items useable by the player via a “use item” button assigned to a keyboard key (“e” if we are using WASD etc). They need to be picked up from the map and have: -A sprite for appearance -A clear functionality-A defined number of uses. -A counter for items used.	UR_ITEMS
FR_INVENTORY	The player has a hotbar of items at the bottom of the screen, with one selected at a time for use. The selected item can be changed by scrolling through with the scroll wheel, or using number hotkeys visible on each hotbar box.	UR_ITEMS
FR_PACIFICATION	The player will “pacify” the geese through the use of items. The pacified state will be represented by a unique sprite.	UR_NONVIOLENCE
FR_MAP	The map is a plane that must feature items, buildings, a win condition and hidden events for the player to locate.	UR_ITEMS UR_EVENTS. UR_MAP
FR_EVENTS	Defined triggers for a set of events that change the game state in some way, and a counter for how many have been triggered.	UR_EVENTS
FR_MOVEMENT	The player needs to be able to move in each cardinal direction and diagonal.	UR_PLAYERCHARACTER
FR_GEESEHORDE	Goose npcs that: -Physically obstruct & damage the player -Respond to items and events (mainly by dispersing)	UR_GEESE

FR_TUTORIAL	There should be a popup screen accessible by other systems with screenshots and diagrams explaining the controls of the game.	UR_TUTORIAL UR_ACCESSIBILITY
FR_WINCONDITION	There needs to be a point where upon reached by the player, the game ends when interacting with it in some form and grants the player a substantial score bonus- just track if it was triggered or not for now. It should be the place blocked off most by the geese and requires clever use of items or movement to get to. Customer wants game to end upon reaching win condition.	UR_WINCONDITION
FR_STARTMENU	Upon booting the game or restarting a run/exiting to menu, the game should have a start menu where difficulty can be selected, a tutorial button like in the pause menu, the game can be started, and it can be quit to desktop	UR_PAUSING UR_DIFFICULTIES UR_TUTORIAL
FR_PLAYERHEALTH	The player should have a health bar that when it is reduced to 0 via damage from goose or other sources, they immediately lose regardless of timer progress.	UR_PLAYERCHARACTER
FR_DIFFICULTY	The game should have a universal variable of what difficulty the current run is set to, set at the start of the run in the main menu. It should affect how hard the game is by increasing values and adding or taking away features (e.g pouncing geese, no pausing) The default value should be the easiest, having pausing and no new features.	UR_DIFFICULTY, UR_LOWDIFF
FR_GAMEEND	The game should have an end state when the game ends after 5 minutes or the player otherwise “dies” from damage. The item, event counters and if they won should be displayed. There should be a button back to the main menu, one to restart the run with the current settings, and one to quit the game.	UR_SCORE, UR_GAMETIME
FR_BARRIER	There should be clearly displayed barriers that the player and geese cannot walk through.	UR_MAP, UR_ACCESSIBILITY
FR_ITEMPOPUPS	Item pop-ups should occur when a player picks up an item which displays the name of the item and what it does.	UR_ITEMS

Quality requirements

ID	Description	User requirements	Fit Criteria
QR_GEESEHINDERANCE	Geese should be blocking the player frequently.	UR_GEESE	There should be geese blocking the user’s path 90% of the time without player input.
QR_ITEMLOCATION	The game should not spawn multiple items within a set, small location(ie grouped together)	UR_ITEMS	No two items should spawn on the same game screen.
QR_EASY	Players on the default difficulty should be able to breeze through it easily.	UR_DIFFICULTIES UR_LOWDIFF	50% of players can beat the game on their first try.
QR_INCLUSIVITY	Players should have multiple different ways of understanding an event has happened. Both visuals and sound to include those with disabilities.	UR_ACCESSIBILTY	80% of players should be able to parse a given event.
QR_UNDERSTANDING	Players should always know what to do through the use of tutorials.	UR_TUTORIAL	80% of players should be able to immediately grasp how to play.